`niram-css v0.2.0`

`niram-css`[1] brings standard CSS color names to Typst.

While Typst natively ships with 18 predefined colors, using anything outside that set usually means hunting down a hex code. `niram-css` eliminates this friction by giving you access to the standard CSS color names. Instead of writing `#c71585`, you can just write `mediumvioletred`.

Key features:

- **Complete CSS Color Support:** Access all 147 standard CSS color names (see Table 1).[2]
- **Flexible Naming Convention:** Write color names in whatever form feels most natural to you. Inputs are automatically normalized, meaning `mediumvioletred`, `medium violet red`, `Medium Violet Red`, `medium-violet-red`, `medium_violet_red`, `mediumVioletRed`, `MediumVioletRed`, `MEDIUM_VIOLET_RED`, and `MEDIUMVIOLETRED` will all resolve to the same color.

Requirements

This package requires Typst 0.13.0 or higher.

Getting Started

Import the package at the top of your Typst document:

```
#import "@preview/niram-css:0.2.0": *
```

Then, use the `css` function anywhere a color value is expected:

```
This is #text(fill: css("mediumvioletred"))[mediumvioletred] text.
```

This is mediumvioletred text.

That is all it takes to start using CSS color names in your document.

Usage

Function Reference

`css`

Resolves a CSS color name to its native Typst `color` value.

The input is automatically normalized before lookup by stripping spaces, hyphens, and underscores, and converting to lowercase. This allows you to write color names using whichever convention fits your workflow.

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[1] In Tamil, *niram* (நிறம், /ɲɪrʉm/) means "color"—a fitting name for a coloring package.

[2] Also available as an HTML page you can view in a browser.

For example, all of the naming conventions below will resolve to `rgb("#c71585")`:

Naming Convention	Example	Common in
lowercase	`mediumvioletred`	CSS / SVG
Spaces	`medium violet red`	X11 / GIMP
Title Case	`Medium Violet Red`	UI Labels / Figma
kebab-case	`medium-violet-red`	SASS / Less / Tailwind
snake_case	`medium_violet_red`	Python / R
camelCase	`mediumVioletRed`	JavaScript / JSON / Swift
PascalCase	`MediumVioletRed`	X11 / .NET / LaTeX xcolor
SCREAMING_SNAKE_CASE	`MEDIUM_VIOLET_RED`	Bash / C constants
UPPERCASE	`MEDIUMVIOLETRED`	JavaFX

Parameters

```
css(color-name: str) -> color
```

color-name `str`

The CSS color name in any supported naming convention.

Examples

Since `niram-css` produces native Typst colors, they work anywhere Typst accepts one—in fills, strokes, underlines, shapes, and color functions like `lighten` or `darken`. The examples below illustrate this in practice.

1. Filling a rectangle with a CSS color.

```
#import "@preview/niram-css:0.2.0": *

#rect(width: 100%, height: auto, fill: css("darkorchid"))
```

2. CSS colors written in lowercase and PascalCase.

```
#import "@preview/niram-css:0.2.0": *

#grid(
  columns: 2,
  rows: 2,
  row-gutter: 10pt,
  column-gutter: 10pt,
  [saddlebrown],
```

```
  [#box(width: 30mm, height: 1em, fill: css("saddlebrown"))],
  [SaddleBrown],
  [#box(width: 30mm, height: 1em, fill: css("SaddleBrown"))],
)
```

saddlebrown

SaddleBrown

3. Lightening and darkening a CSS color with Typst's `lighten` and `darken` functions.

```
#import "@preview/niram-css:0.2.0": *

#grid(
  columns: 2,
  rows: 3,
  row-gutter: 10pt,
  column-gutter: 10pt,
  [crimson],
  [#box(width: 30mm, height: 1em, fill: css("crimson"))],
  [crimson lightened 40%],
  [#box(width: 30mm, height: 1em, fill: css("crimson").lighten(40%))],
  [crimson darkened 40%],
  [#box(width: 30mm, height: 1em, fill: css("crimson").darken(40%))],
)
```

crimson

crimson lightened 40%

crimson darkened 40%

4. Typst's built-in `green` differs from the standard CSS `green`. Access the CSS version with `css("green")`.[3]

```
#import "@preview/niram-css:0.2.0": *

#grid(
  columns: 3,
  rows: 2,
  row-gutter: 10pt,
  column-gutter: 10pt,
  [Typst green],
  [#color.green.to-hex()],
  [#box(width: 30mm, height: 1em, fill: green)],
  [CSS green],
  [#css("green").to-hex()],
  [#box(width: 30mm, height: 1em, fill: css("green"))],
)
```

Typst green #2ecc40

[3] See Same Name, Different Hex Value for why they differ.

CSS green #008000

5. Using CSS colors to underline text.

```
#import "@preview/niram-css:0.2.0": *

Thin #underline(stroke: css("darkblue"), offset: 2pt)[underline.]

Double #underline(
  stroke: (paint: css("darkblue"), thickness: 0.5pt),
  offset: 2pt,
)[
  #underline(
    stroke: (paint: css("darkblue"), thickness: 0.5pt),
    offset: 5pt,
  )[underline.]
]

Thick #underline(stroke: (paint: css("darkblue"), thickness: 2pt), offset: 2pt)
[underline.]
```

Thin underline.

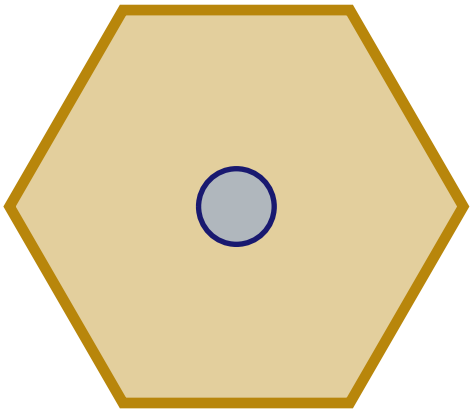
Double underline.

Thick underline.

6. Using CSS colors to draw shapes.

```
#import "@preview/niram-css:0.2.0": *

#align(center)[
  #block[
    #polygon.regular(
      fill: css("darkgoldenrod").lighten(60%),
      stroke: (paint: css("darkgoldenrod"), thickness: 4pt),
      size: 60mm,
      vertices: 6,
    )
    #place(center + horizon)[
      #circle(
        radius: 5mm,
        stroke: (paint: css("midnightblue"), thickness: 2pt),
        fill: css("cornflowerblue").transparentize(60%),
      )
    ]
  ]
]
```



Caveats

Same Name, Different Hex Value

Typst provides a set of 18 predefined colors. 17 of these borrow their names directly from the CSS Level 2 specification. The remaining color, `eastern`, is a proprietary addition unique to Typst. It was added specifically to provide built-in access to Typst's brand color and has no equivalent in the CSS specification.

However, it is important to distinguish between color names and color values. While 17 of Typst's predefined colors share their names with standard CSS colors, their actual hex values are not the same.

According to Typst's source, their default hex values are adapted from the `colors.css` project by designer Adam Morse. Morse deliberately took familiar CSS color names and mapped them to softer, perceptually balanced hex values for better design aesthetics. Because Typst adopted Morse's palette, its default colors are noticeably different from the standard CSS colors you might be expecting, as shown in Table 4.

X11 Gray Equivalents in Typst

`niram-css` covers the CSS standard, which omits X11's numbered color families (e.g., `red1–red4`, `blue1–blue4`). These are rarely used today.

Numbered grays (`gray0–gray100`) are an exception: X11 defines `grayN` as N% brightness, so they can be reproduced exactly with Typst's `lighten` function. For example, `gray42` is `black.lighten(42%)`.

The History of the CSS Color Standard

X11 Colors

The X11 color names, developed in the 1980s, were defined in a file named `rgb.txt` shipped with every X11 installation. The file mapped over 400 color names to specific RGB values. This list became the de facto color naming convention on Unix-based systems, but was never formally specified by Xlib or the X11 protocol. Because `rgb.txt` was vendor-editable, different X server implementations shipped variations of the list.

The 17 Foundational Color Names of the Early Web

The HTML 3.2 specification of 1996 introduced 16 color names, drawn from the Windows VGA palette. Known as the 16 named colors.

Thanks to this standard, these 16 color names could be used consistently in documents across the web, without needing to specify hex values.

CSS Level 1, introduced in 1996, adopted these same 16 colors, while CSS Level 2 (1998) expanded the list to 17 with the addition of `orange`. These colors along with their hex values are listed in Table 2.

CSS/SVG Colors

The SVG 1.1 specification of 2011 and the CSS Color Module Level 3 of 2022 standardized 147 color names, known as the "recognized color keyword names." They comprise the original 16 HTML/VGA named colors, `orange`, and an additional 130 color names drawn from the X11 set.

This standardized list—the basis for this package—provides a consistent color naming scheme across the web and has remained stable since its adoption. Alex Sexton explains how the CSS colors got their idiosyncratic names in an informative but lighthearted video on YouTube.

While all 147 CSS color names are distinct, only 138 resolve to unique hex values: 9 hex values are each shared by 2 names. These 9 pairs are listed in Table 3.

CSS Named Color Reference

Table 1: All 147 CSS named colors and their hex values, sorted alphabetically.

Color Name	Hex Value	Stripe
aliceblue	#f0f8ff	
antiquewhite	#faebd7	
aqua	#00ffff	
aquamarine	#7fffd4	
azure	#f0ffff	
beige	#f5f5dc	
bisque	#ffe4c4	
black	#000000	
blanchedalmond	#ffebcd	
blue	#0000ff	
blueviolet	#8a2be2	
brown	#a52a2a	
burlywood	#deb887	
cadetblue	#5f9ea0	
chartreuse	#7fff00	
chocolate	#d2691e	
coral	#ff7f50	
cornflowerblue	#6495ed	

Color Name	Hex Value	Stripe
cornsilk	#fff8dc	
crimson	#dc143c	
cyan	#00ffff	
darkblue	#00008b	
darkcyan	#008b8b	
darkgoldenrod	#b8860b	
darkgray	#a9a9a9	
darkgreen	#006400	
darkgrey	#a9a9a9	
darkkhaki	#bdb76b	
darkmagenta	#8b008b	
darkolivegreen	#556b2f	
darkorange	#ff8c00	
darkorchid	#9932cc	
darkred	#8b0000	
darksalmon	#e9967a	
darkseagreen	#8fbc8f	
darkslateblue	#483d8b	
darkslategray	#2f4f4f	

Color Name	Hex Value	Stripe
darkslategrey	#2f4f4f	
darkturquoise	#00ced1	
darkviolet	#9400d3	
deeppink	#ff1493	
deepskyblue	#00bfff	
dimgray	#696969	
dimgrey	#696969	
dodgerblue	#1e90ff	
firebrick	#b22222	
floralwhite	#fffaf0	
forestgreen	#228b22	
fuchsia	#ff00ff	
gainsboro	#dcdcdc	
ghostwhite	#f8f8ff	
gold	#ffd700	
goldenrod	#daa520	
gray	#808080	
green	#008000	
greenyellow	#adff2f	

Color Name	Hex Value	Stripe
grey	#808080	
honeydew	#f0fff0	
hotpink	#ff69b4	
indianred	#cd5c5c	
indigo	#4b0082	
ivory	#fffff0	
khaki	#f0e68c	
lavender	#e6e6fa	
lavenderblush	#fff0f5	
lawngreen	#7cfc00	
lemonchiffon	#fffacd	
lightblue	#add8e6	
lightcoral	#f08080	
lightcyan	#e0ffff	
lightgoldenrodyellow	#fafad2	
lightgray	#d3d3d3	
lightgreen	#90ee90	
lightgrey	#d3d3d3	
lightpink	#ffb6c1	

Color Name	Hex Value	Stripe
lightsalmon	#ffa07a	
lightseagreen	#20b2aa	
lightskyblue	#87cefa	
lightslategray	#778899	
lightslategrey	#778899	
lightsteelblue	#b0c4de	
lightyellow	#ffffe0	
lime	#00ff00	
limegreen	#32cd32	
linen	#faf0e6	
magenta	#ff00ff	
maroon	#800000	
mediumaquamarine	#66cdaa	
mediumblue	#0000cd	
mediumorchid	#ba55d3	
mediumpurple	#9370db	
mediumseagreen	#3cb371	
mediumslateblue	#7b68ee	
mediumspringgreen	#00fa9a	

Color Name	Hex Value	Stripe
mediumturquoise	#48d1cc	
mediumvioletred	#c71585	
midnightblue	#191970	
mintcream	#f5fffa	
mistyrose	#ffe4e1	
moccasin	#ffe4b5	
navajowhite	#ffdead	
navy	#000080	
oldlace	#fdf5e6	
olive	#808000	
olivedrab	#6b8e23	
orange	#ffa500	
orangered	#ff4500	
orchid	#da70d6	
palegoldenrod	#eee8aa	
palegreen	#98fb98	
paleturquoise	#afeeee	
palevioletred	#db7093	
papayawhip	#ffefd5	

Color Name	Hex Value	Stripe
peachpuff	#ffdab9	
peru	#cd853f	
pink	#ffc0cb	
plum	#dda0dd	
powderblue	#b0e0e6	
purple	#800080	
red	#ff0000	
rosybrown	#bc8f8f	
royalblue	#4169e1	
saddlebrown	#8b4513	
salmon	#fa8072	
sandybrown	#f4a460	
seagreen	#2e8b57	
seashell	#fff5ee	
sienna	#a0522d	
silver	#c0c0c0	
skyblue	#87ceeb	
slateblue	#6a5acd	
slategray	#708090	

Color Name	Hex Value	Stripe
slategrey	#708090	
snow	#fffafa	
springgreen	#00ff7f	
steelblue	#4682b4	
tan	#d2b48c	
teal	#008080	
thistle	#d8bfd8	
tomato	#ff6347	
turquoise	#40e0d0	
violet	#ee82ee	
wheat	#f5deb3	
white	#ffffff	
whitesmoke	#f5f5f5	
yellow	#ffff00	
yellowgreen	#9acd32	

The 17 Foundational Colors

Table 2: The 16 HTML 3.2 named colors and their hex values.

Color Name	Hex Value	Stripe
aqua	#00ffff	
black	#000000	
blue	#0000ff	
fuchsia	#ff00ff	
gray	#808080	
green	#008000	
lime	#00ff00	
maroon	#800000	
navy	#000080	
olive	#808000	
orange	#ffa500	
purple	#800080	
red	#ff0000	
silver	#c0c0c0	
teal	#008080	
white	#ffffff	
yellow	#ffff00	

CSS Colors Sharing Hex Values

Table 3: CSS color names that share the same hex value.

Color Names	Hex Value	Stripe
aqua cyan	#00ffff	
darkgray darkgrey	#a9a9a9	
darkslategray darkslategrey	#2f4f4f	
dimgray dimgrey	#696969	
fuchsia magenta	#ff00ff	
gray grey	#808080	
lightgray lightgrey	#d3d3d3	
lightslategray lightslategrey	#778899	
slategray slategrey	#708090	

Typst's Predefined Colors versus CSS Colors

Table 4: Typst's predefined colors and their CSS counterparts.

Color Name	Typst Hex Value	Stripe	CSS Hex Value	Stripe
aqua	#7fdbff		#00ffff	
black	#000000		#000000	
blue	#0074d9		#0000ff	
eastern	#239dad		—	—
fuchsia	#f012be		#ff00ff	
gray	#aaaaaa		#808080	
green	#2ecc40		#008000	
lime	#01ff70		#00ff00	
maroon	#85144b		#800000	
navy	#001f3f		#000080	
olive	#3d9970		#808000	
orange	#ff851b		#ffa500	
purple	#b10dc9		#800080	
red	#ff4136		#ff0000	
silver	#dddddd		#c0c0c0	
teal	#39cccc		#008080	
white	#ffffff		#ffffff	
yellow	#ffdc00		#ffff00	